

Scaling Human-AI Quality Control in Semiconductor Manufacturing: A Dual-Legibility Visual Analytics Approach

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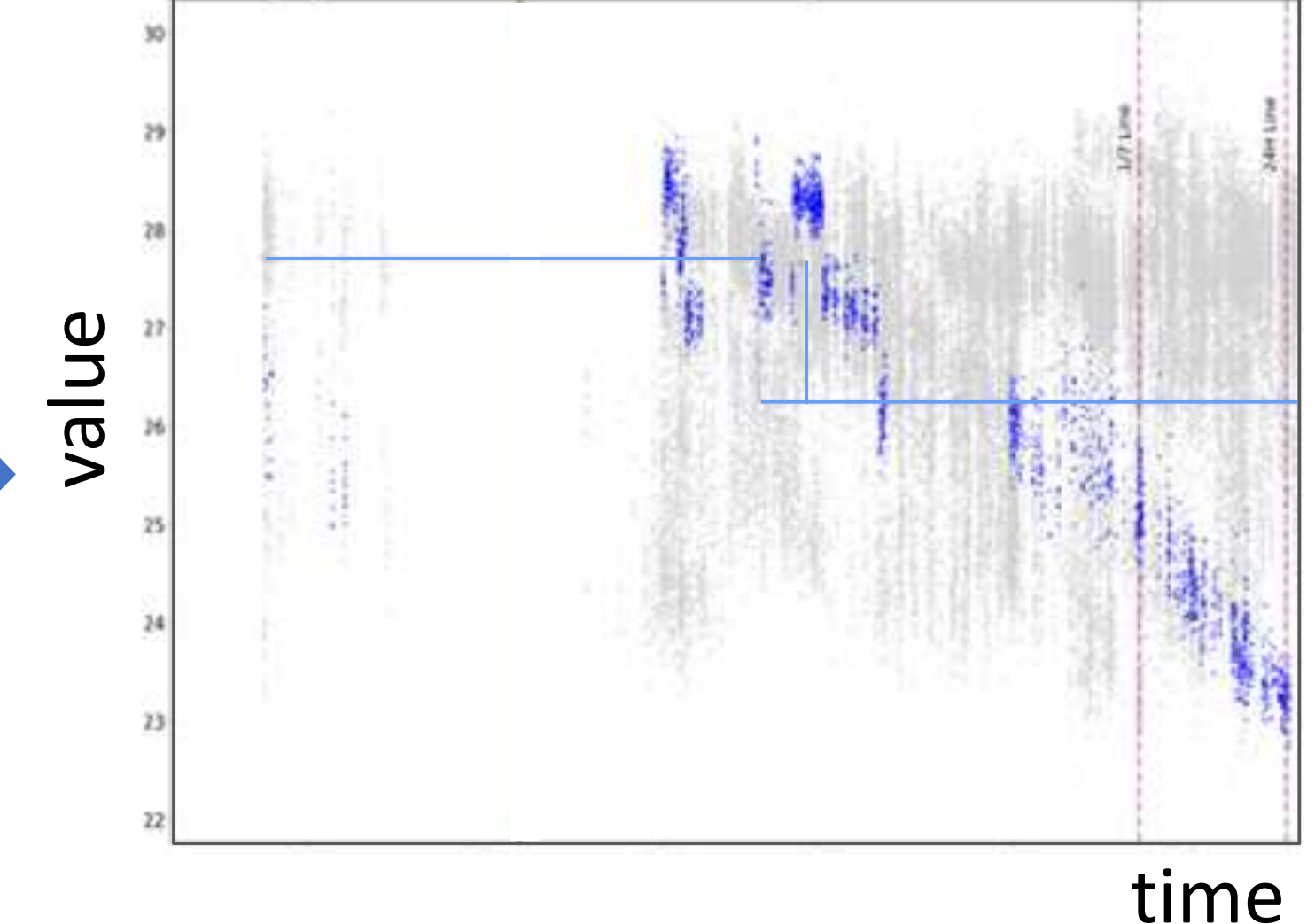
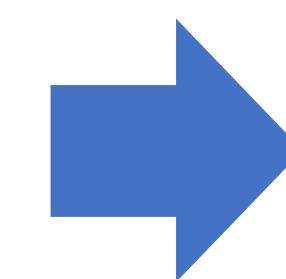
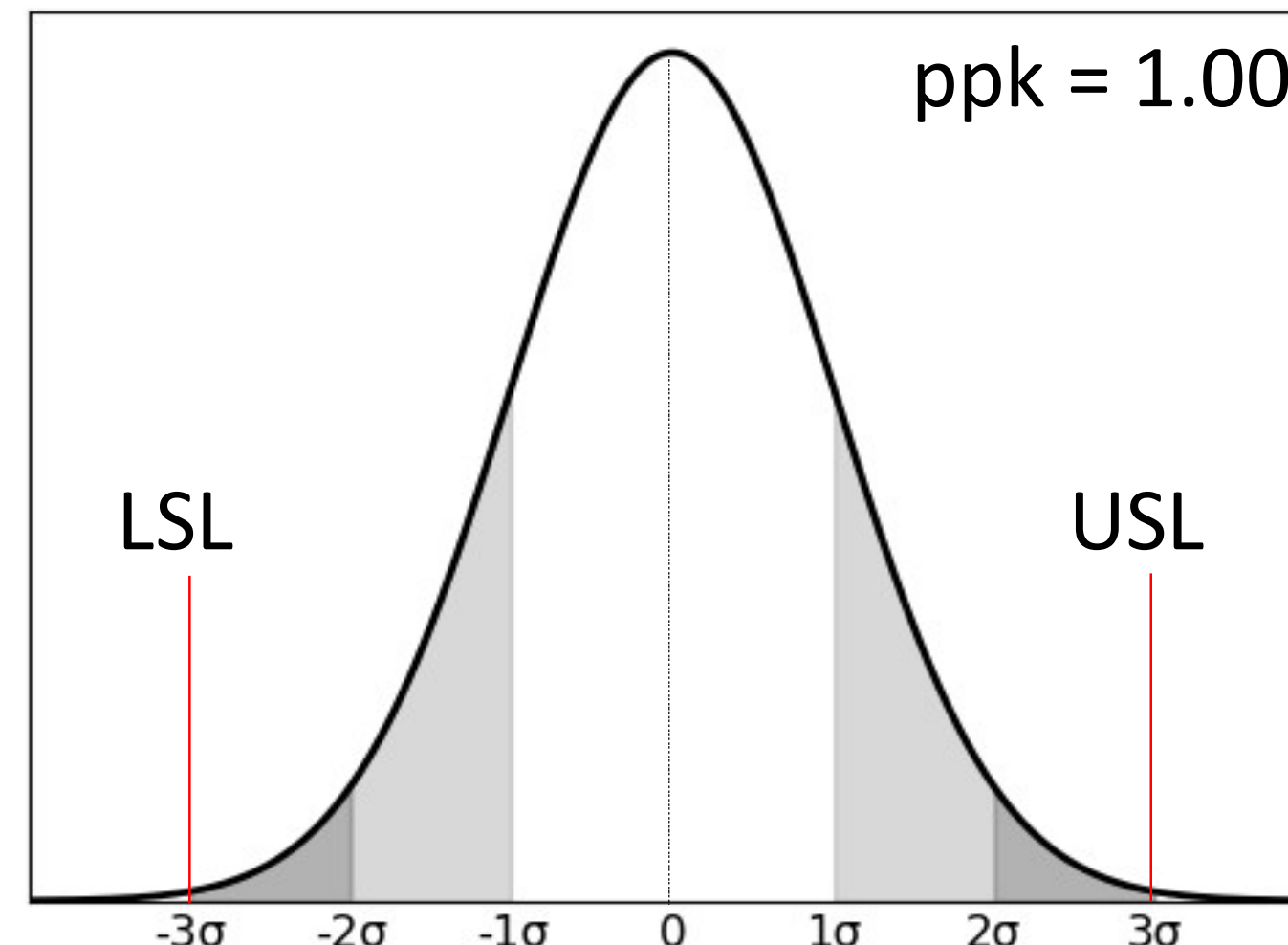
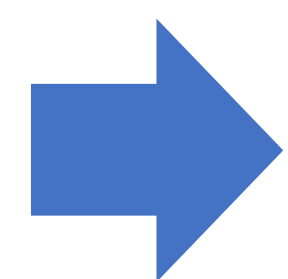
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The Challenge: Scaling Quality Control



Time	LotID	Slot	EQP_ch.	Value
3/18 13:20	Lot_id1	22	EQP1-2	27.36
3/18 16:18	Lot_id2	15	EQP2-1	26.58
3/18 22:04	Lot_id3	9	EQP1-3	28.69
3/19 01:32	Lot_id4	5	EQP2-2	24.32
3/19 03:13	Lot_id5	19	EQP1-1	25.94
3/19 06:11	Lot_id6	13	EQP2-3	24.19



Semiconductor equipment
Generates massive volume of complex data.

Statistical Process Control
Struggles to detect subtle, multivariate anomalies.

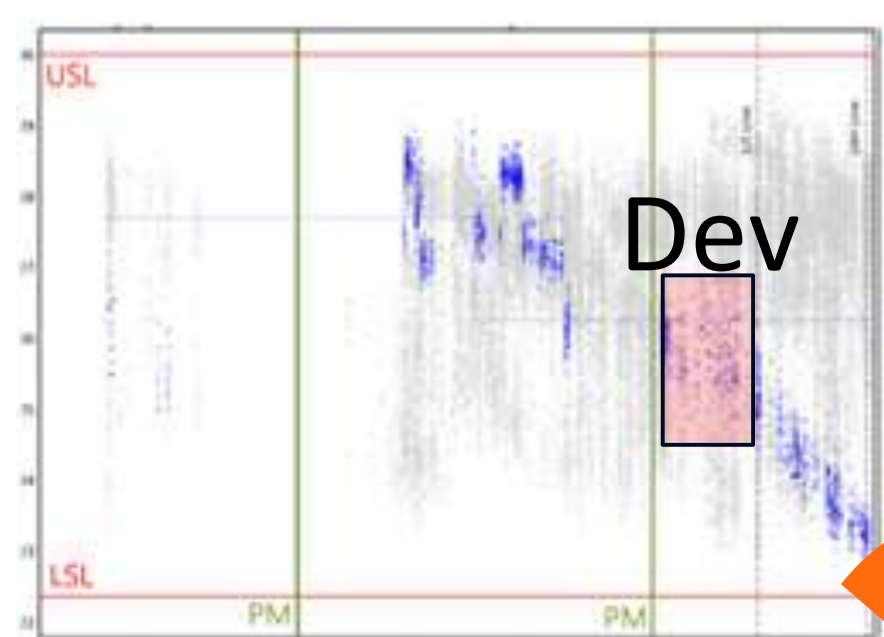
Manual chart review
Creates a bottleneck with experts' review → **The need for AD**
*Anomaly Detection

「The problems of early AI」

- ✓ **Excessive false alarms:** hampered by low precision (only 53.2%), leading to unreliable alerts.
- ✓ **Unexplainable judgments:** lacking insight into "where" and "why," blocking root-cause analysis → **The need for OD**
*Object Detection

Visual Analytics Loop (VAL) based on the dual-legibility chart

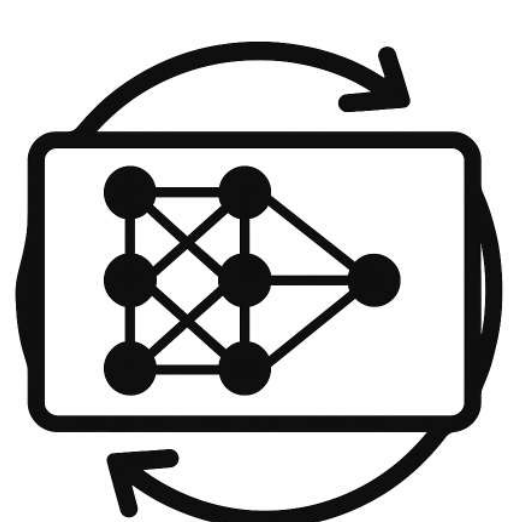
1. AI Prediction



2. Expert Refinement



4. Model Retraining

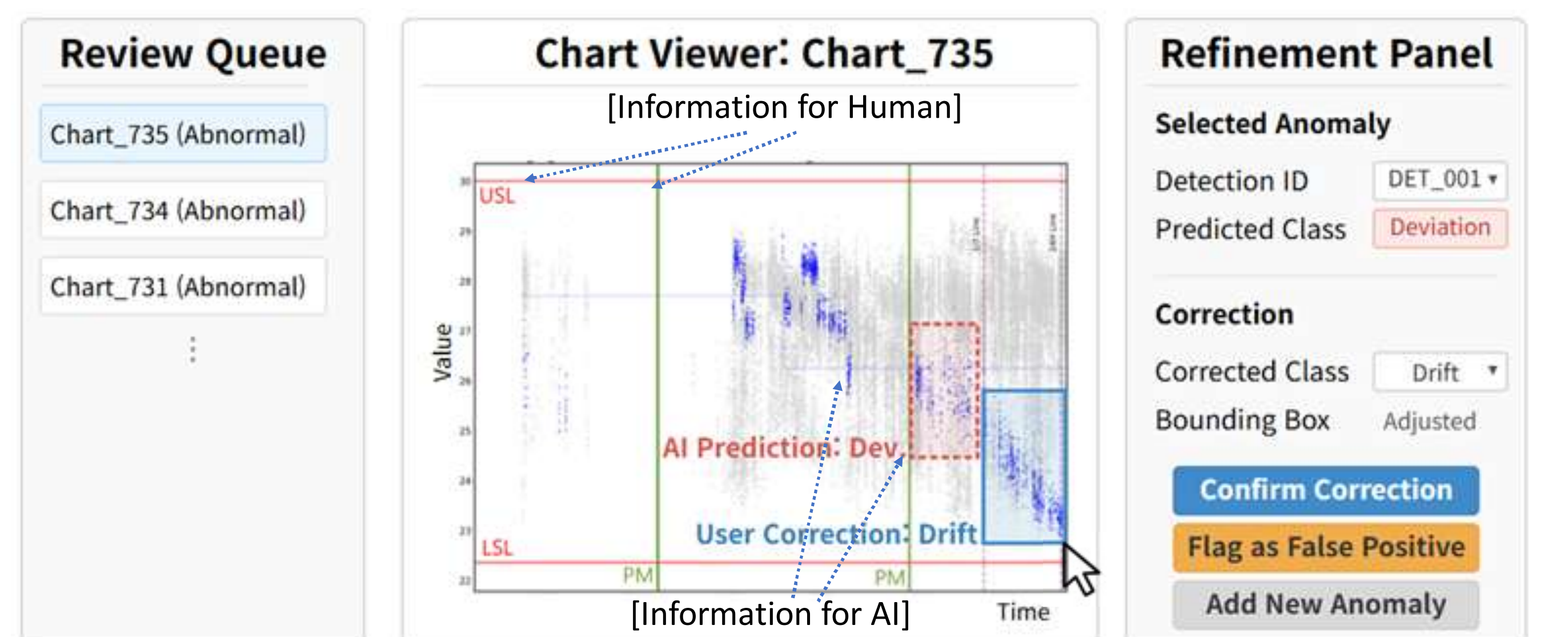


3. Feedback Generation



【Interactive label refinement interface】

Designed following Shneiderman's visual information-seeking mantra [2]



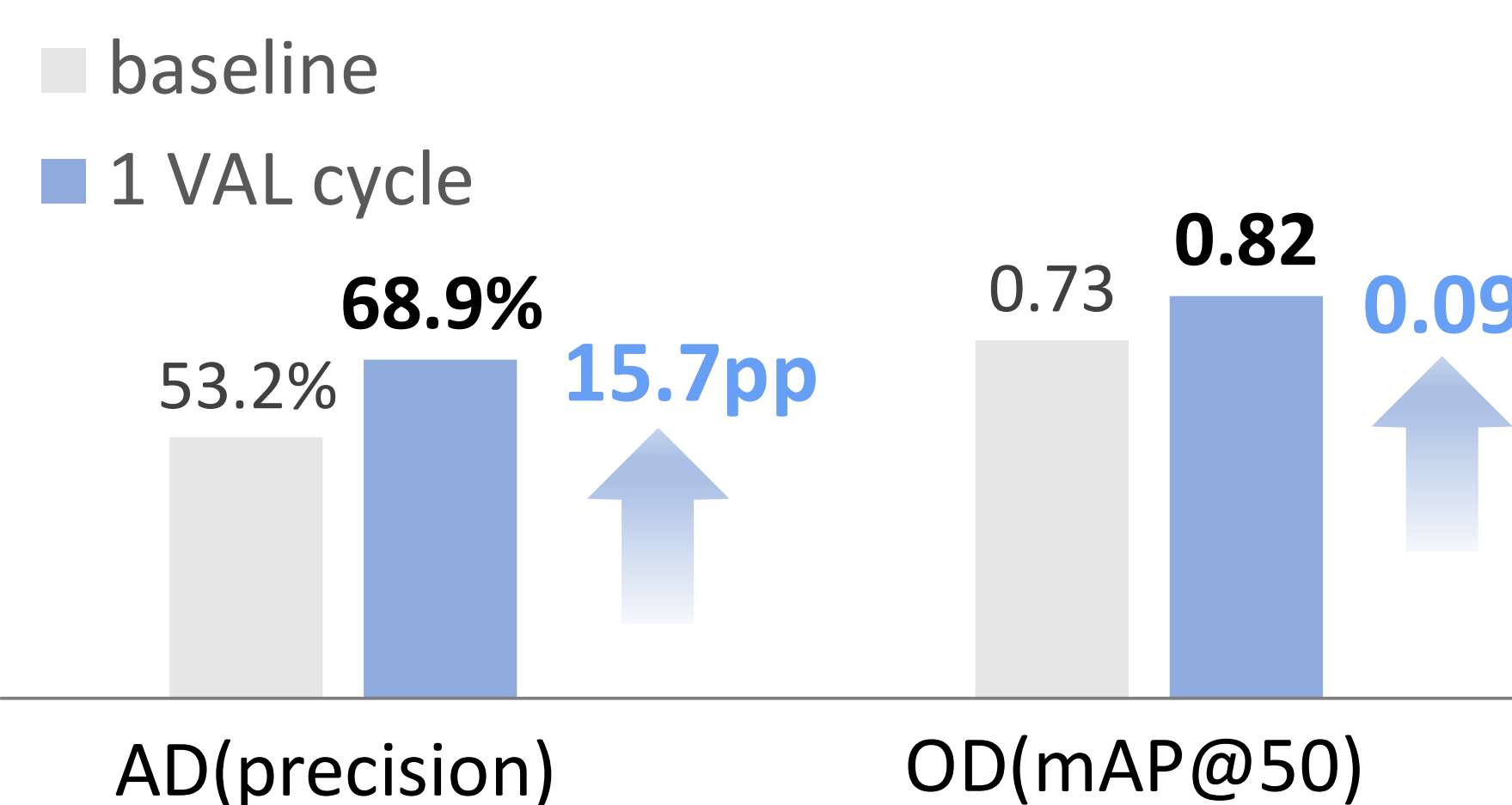
Engineers can **directly correct** AI's mistakes on the chart

「The Dual-Legibility chart: a shared language」

- ✓ **For human:** Critical engineering context (SPC limits, PM)
- ✓ **For AI:** Structured labels (class, bounding box, AD flag)

Inspired by human in the AI loop for visual inspection [1]
The VAL framework: A closed-loop that uses expert edits to continuously retrain and improve the AI.

Contributions & Future Directions



“So intuitive and trustworthy.”

— Engineers, on the direct interaction via the dual-legibility chart.

Contribution

- **Dual-legibility chart:** a shared human-AI visual interface.
- **VAL workflow:** enables rapid model refinement using expert edits.
- **Production validation:** proven effective with performance gains.

Future work

- **Generalizability:** currently limited, plan to scale for broader use.
- **Usability:** tested on experts, plan to optimize for wider user base.

Reference

J. Rožanec et al., “Human in the AI loop via xAI and active learning for visual inspection.”, B. Shneiderman, “The eyes have it: A task by data type taxonomy for information visualizations.”